

Jonathan J. Yang

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EDUCATION

The University of Texas at Austin

Austin, TX | May 2028

Bachelor of Science in Computer Science, **Turing Scholars Honors**

GPA: 3.9/4.0

Coursework (*H=Honors*): *Machine Learning (H)*, *Algorithms (H)*, *Data Structures (H)*, *Discrete Math (H)*, *Vector Calculus (H)*, *Microprocessor Memory Systems (H)*, *Operating Systems (H)*, *Computer Architecture (H)*, Generative Visual Computing, Game Theory, Probability, Number Theory, Matrices, Quantum Physics

TECHNICAL SKILLS

Proficient Languages: C/C++, Python, GoLang, Java, SQL, Javascript/Typescript, Verilog, R, Terraform

Systems/Libraries: AWS, OCI, Docker, Git, Linux, Jira, React, Pytorch, CUDA

WORK EXPERIENCE

Software Developer Intern @ Oracle Cloud Infrastructure (OCI)

Nasville, TN | May 2026 – Aug 2026

- Engineered and deployed a responsive full-stack monitoring application for the CI/CD build service of the Customer Compute Control Plane (CCCP) to surface flaky failures and details.
- Optimized to provide a preview of the >200 TiB of archive data using <1 GiB memory resources; user latency is <1s.

OTHER EXPERIENCE

Student Researcher @ UT LDOS Lab

Austin, TX | Nov 2025 – Present

- Extracted traces from various Linux kernel subsystems via eBPF probes and replayed for page cache prefetching.
- Researching context-aware prefetching at the hypervisor level for concurrent compute workloads on distributed storage systems by engineering simulations on well-known benchmarks (e.g. YCSB).

Founding Member @ Apovo (startup acquired by Livestrong)

Austin, TX | Nov 2024 – Jul 2026

- Built a medical-focused full-stack application with HIPAA compliance, secure identification and authentication, and good uptime by implementing security protocols, MFA, and blacklisting (blocked 10k bot requests/day).
- Engineered a recommendation system combining matrix factorization and content-based matching; improving precision from .2, .0 to .7, .2 for queries and recommendations on cold start, respectively.
- Optimized TTFB from 20s to 1s through caching, batch updates, model distillation, and progressive rendering.

Software Developer Intern @ Kent PLC

Dubai, UAE (Remote) | May 2025 – August 2025

- Developed assignment automation in ERP system to escalate tickets and leave requests to proper supervisors in Microsoft SQL database.

SELECTED PROJECTS

Programming Language Project (github.com/Jzhyang1/property-lang)

Sprint 2026

- Implemented a functional-language inspired LLVM frontend with JIT-compilation, static module generation, Python integration, and safe AI code-generation semantics.

Generative Visual Computing Projects

Fall 2025

- Image generation through DCGAN, WGAN, DDPM (with DDIM sampling) on MNIST and CIFAR10.
- Motion generation via AutoEncoder on CMU Mocap.
- Texture synthesis through nonparametric pixel growth and patch-quilting methods.

High Throughput Kernel Lock-Free Scheduling Queue

Fall 2025

- Designed a custom lock-free queue optimized for a preemptive scheduler on a multicore operating systems; achieved 75% throughput improvement over Michael-Scott queue for high contention cases.

Computer Architecture Class Project

Spring 2025

- Implemented pipelined RISC CPU (loads, stalls, arithmetic, branch prediction) in Verilog.
- Expanded project with a team to lower onto an FPGA; added peripherals, tilemap, and implemented Conway's game of Life and Snake on live system.